

CODE	MATHEMATICAL	CATEGORY	L	T	P	CREDIT
221TCS003	FOUNDATIONS FOR DATA SCIENCE	Program Core 1	3	0	0	3

Preamble:

This course is intended to provide the learners with an outlook on applying concepts in linear algebra in the fields of data science, machine learning, and artificial intelligence. This course helps the learners to acquire the skills to implement the concepts in MATLAB/Python and then apply linear algebra concepts to real datasets. Also, this course discusses the Challenges of applying the acquired knowledge in different Optimization and Linear Algebra concepts toward the inference and prediction stages of Data Analytics.

Course Outcomes: After the completion of the course, the student will be able to

CO 1	Analyse the fundamentals of linear algebra and calculus, and other mathematical concepts for Artificial Intelligence, Machine Learning, and Data Science (Cognitive knowledge level: Analyse)
CO 2	Apply the knowledge acquired in different optimization and linear algebra concepts towards the inference and prediction stages of data analytics. (Cognitive knowledge level: Apply)
CO 3	Implement linear algebra concepts in scientific programming languages (MATLAB, Python) (Cognitive knowledge level: Apply)
CO 4	Apply eigenvectors and SVD for Dimensionality reduction (Cognitive knowledge level: Apply)
CO 5	Design, Develop, implement, and Present innovative Ideas on the application of linear algebra for Data Science, Machine learning, and Artificial Intelligence (Cognitive Knowledge Level: create)

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1: An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2: An ability to communicate effectively and write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3: An ability to demonstrate mastery over the area as per the program's specialization. The mastery should be at a level higher than the requirements in the appropriate bachelor's program

PO4: An ability to apply stream knowledge to design or develop solutions for real-world problems by following the standards

PO5: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tools to model, analyze and solve practical engineering problems.

PO6: An ability to engage in life-long learning for the design and development related to the stream-related problems taking into consideration sustainability, societal, ethical, and environmental aspects

PO7: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	☑	☑	☑	☑	☑	☑	
CO 2	☑	☑	☑	☑	☑	☑	
CO 3	☑	☑	☑	☑	☑	☑	
CO 4	☑	☑	☑	☑	☑	☑	
CO 5	☑	☑	☑	☑	☑	☑	☑

Assessment Pattern

Bloom's Category	End Semester Examination
Understand	
Apply	60-80%
Analyze	20-40%
Evaluate	Can be done through Assignments
Create	Can be done through Assignments

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern:

The evaluation shall only be based on application, analysis or design-based questions (for both internal and end-semester examinations).

Continuous Internal Evaluation : 40 marks

Micro project/Course based project : 20 marks

Course based task/Seminar/Quiz : 10 marks

Test paper, 1 no. : 10 marks

The project shall be done individually. Group projects are not permitted.

The test paper shall include a minimum of 80% of the syllabus.

Course-based task/test paper questions shall be useful in testing the knowledge, skills, comprehension, application, analysis, synthesis, evaluation, and understanding of the students.

End Semester Examination Pattern:

The end-semester examination will be conducted by the University. There will be two parts; Part A and Part B. Part A contain 5 numerical questions with 1 question from each module, having 5 marks for each question. (Such questions shall be useful in testing knowledge, skills, comprehension, application, analysis, synthesis, evaluation, and understanding of the students). Students shall answer all questions.

Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions relating to theoretical/practical knowledge, derivations, problem-solving and quantitative evaluation), with a minimum one question from each module of which student should answer any five. Each question can carry 7 marks.

The total duration of the examination will be 150 minutes.

Course Level Assessment Questions

Course Outcome 1 (CO1):

1. Discuss Linear algebra handles large amounts of data,
2. List ten Powerful Applications of Linear Algebra in Data Science
3. 'Support Vector Machine is an application of the concept of Vector Spaces in Linear Algebra'- Investigate
4. Support Vector Machine Classification

Course Outcome 2 (CO2)

1. Write an algorithm for simple linear regression by gradient descent method
2. Implement Gradient Descent for multilinear regression from scratch
3. Elaborate regularized linear regression for model prediction and reduce errors

Course Outcome 3(CO3):

1. Elaborate on the different ways to compute and conceptualize matrix multiplication with examples
2. Implement Vector Hadamard multiplication
3. $w1 = [1\ 3\ 5]$;
4. $w2 = [3\ 4\ 2]$; in MATLAB/ Python

Course Outcome 4 (CO4):

1. Implement the determinant of a matrix product in Python
2. Elucidate the usefulness of Linear Algebra in Dimensionality Reduction
 - i. Principal Component Analysis (PCA)
 - ii. Singular Value Decomposition (SVD)
3. Determine whether the following matrices have a null space. If so, provide basis vector(s) for that null space.

$$a. \begin{pmatrix} 4 & 3 \\ 1 & 1 \\ 0 & 5 \end{pmatrix}$$

$$b. \begin{pmatrix} 3 & 7 & 5 \\ 4 & 1 & 0 \end{pmatrix}$$

Course Outcome 5 (CO5):

1. List the five steps of model fitting
2. Express the average or the mean of a set of numbers as a model and use least squares to fit that model
3. Elaborate on the usage of Linear Algebra in Machine Learning
 - a. Loss functions
 - b. Regularization
 - c. Covariance Matrix

Model Question Paper

QP CODE:

Reg No: _____

Name: _____

PAGES : 4

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

FIRST SEMESTER M. TECH DEGREE EXAMINATION, MONTH & YEAR

Course Code: 221TCS003

Course Name: Mathematical Foundation for Data Science

Max. Marks: 60

Duration: 2.5 Hours

PART A

Answer All Questions. Each Question Carries 5 Marks

1. Explain normal equation. Contrast normal equation and gradient descent. 5
2. Explain overfitting and the method of resolving overfitting 5
3. Make use of the parallel matrix multiplication method to find AB, where 5

$$A = \begin{bmatrix} 2 & 1 & 5 & 3 \\ 0 & 7 & 1 & 6 \\ 9 & 2 & 4 & 4 \\ 3 & 6 & 7 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 6 & 1 & 2 & 3 \\ 4 & 5 & 6 & 5 \\ 1 & 9 & 8 & -8 \\ 4 & 0 & -8 & 5 \end{bmatrix}$$

4. Let $v = \langle 1, 3 \rangle$ and $w = \langle -4, -2 \rangle$. Write v as the sum of two orthogonal vectors, one of which is the projection of $v \rightarrow$ onto w 5
5. Find the SVD for the matrix 5

$$A = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix}$$

(5x5=25)

Part B

(Answer any five questions. Each question carries 7 marks)

6. (a) Write an algorithm for simple linear regression by gradient descent method (7)
7. (a) Implement Gradient Descent for multilinear regression from scratch (7)

8. (a) Elaborate the different types of matrices with examples and implement them using MATLAB or NumPy (7)

- i. Square matrix
- ii. Rectangular matrix
- iii. Symmetric matrix
- iv. Skew-symmetric matrices
- v. Identity matrix
- vi. Zero
- vii. Diagonal matrix
- viii. Triangular matrix
- ix. Augmented
- x. Complex

9. (a) Define rank and a maximum possible rank (2)

- (b) Create a matrix of 10 X 10 with a rank of 4 (use matrix multiplication) (3)

- (c) Generalize the procedure to create any M x N matrix with a rank r (2)

10. (a) Implement the determinant of a matrix product in Python (3)

- (b) Determine whether the following matrices have a null space. If so, provide basis vector(s) for that null space. (4)

$$a. \begin{pmatrix} 4 & 3 \\ 1 & 1 \\ 0 & 5 \end{pmatrix}$$

$$b. \begin{pmatrix} 3 & 1 & 5 \\ 4 & 1 & 0 \end{pmatrix}$$

11. (a) Write MATLAB or Python code to implement the following experiment: (7)

- a. Generate a 2×3 matrix of random numbers.
- b. Compute its SVD.
- c. Compute two eigen decompositions using the matrix and its transpose.
- d. Confirm that the two sets of eigenvalues match, and check whether the eigenvalues match the singular values

Plot the eigenvectors and singular vectors in 2D or 3D (as appropriate) to confirm that SVD and transpose + eigen decomposition produce the same eigenspaces

12. (a) Determine the eigenvalues and the corresponding eigenvectors of the following (7)

$$\text{matrix } A = \begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$$

Syllabus (Emphasis shall be on Problem-solving, implementing concepts, and application- Languages Python/MATLAB)

Module	Contents	hours
I	GRADIENT DESCENT AND REGULARIZATION Gradient Descent, Intuitions, Gradient Descent for a Regression algorithm, Multiple Features, Gradient Descent on Multiple Features, Practice on Gradient Descent for Polynomial Regression, Normal Equation	8
II	LINEAR ALGEBRA-VECTORS: Vectors: geometry and algebra, Vector addition, and subtraction, Vector-scalar multiplication, Dot product geometry, Vector orthogonality, Cauchy-Schwarz inequality, Vector Hadamard multiplication, cross product, unit vectors, VECTOR SPACES: Dimensions, and fields in linear algebra, Subspaces, Subspaces vs. subsets, Span, Linear independence, Basis MATRICES: introduction: dimensionality, Matrix operations, Matrix-scalar multiplication, Implementation, Transpose, Complex matrices, Addition, equality, transpose, Diagonal, and trace,	7
III	MATRICES:- MATRIX MULTIPLICATION -Introduction, matrix multiplication by layering, Multiplication with diagonals, Matrix-vector multiplication, Symmetric matrices, multiply symmetric matrices Hadamard Multiplication, asymmetry Index, Code challenge, RANK -concepts, Maximum possible rank, Computing rank, Rank and scalar multiplication, Rank of added and multiplied matrices, Rank of A & A^T , $A^T A$, AA^T , random matrices, Boosting rank by shifting, rank difficulties, rank, and span, Code challenge: MATRIX SPACES: Column space and Row space of a matrix (A & AA^T), Null space of a matrix, orthogonal subspaces, Dimensions of column/row/null spaces, Example of the four subspaces, code challenge	7
IV	DETERMINANTS, PROJECTIONS & ORTHOGONALIZATION- DETERMINANT -Determinant, Determinant of a 2x2 matrix, Determinant of a 3x3 matrix, characteristic polynomial, the full procedure, determinant of triangles, determinant and row reduction, determinant and scalar multiplication, theory vs practice, Code challenge MATRIX INVERSE: Concept and applications, Inverse of a Diagonal matrix, Inverse of a 2x2 matrix, The MCA algorithm to compute the inverse, Computing the inverse via row reduction, Left inverse and right inverse, Pseudo-inverse, Code challenge PROJECTIONS, AND ORTHOGONALIZATIONS: Projections in R^2 , Projections in R^N , Orthogonal and parallel vector components, Orthogonal matrices, Gram-Schmidt procedure, QR decomposition, Inverse via QR Decomposition, Code challenge LEAST SQUARES FOR MODEL-FITTING IN STATISTICS: Introduction, Least squares via left inverse, Least squares via orthogonal projection, Least-squares via row-reduction, Model-predicted values, and residuals, Least-squares applications, Code challenge	10
V	DIMENSIONALITY REDUCTION: EIGEN DECOMPOSITION- Eigenvalues, eigenvectors, Eigen decomposition, Diagonalization, Matrix powers via diagonalization, Distinct and repeated eigenvalues, symmetric	8

	matrices, Eigen layers of a matrix, Eigen decomposition of singular matrices, Matrix powers and Inverse, Generalized eigen decomposition, Code challenges SINGULAR VALUE DECOMPOSITION (SVD) : Singular value decomposition, Computing the SVD, singular values and eigenvalues, Symmetric Matrices, SVD and the four subspaces, SVD, and matrix rank, Spectral theory of matrices, SVD for low-rank approximations, Normalizing singular values, the Condition number of a matrix, SVD and Matrix Inverse, MP pseudo inverse, code challenges	
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Course Plan

S.NO	TOPIC	NO. OF LECTURES
MODULE 1 - GRADIENT DESCENT AND REGULARIZATION-8 hours		
1.1	Gradient Descent, Intuitions,	1
1.2	Gradient Descent for a Regression algorithm.	1
1.3	Gradient Descent for a Regression algorithm.	1
1.4	Multiple Features	1
1.5	Gradient Descent on Multiple Features,	1
1.6	Practice on Gradient Descent	1
1.7	Gradient Descent for Polynomial Regression	1
1.8	Gradient Descent and Normal Equation	1
MODULE 2 LINEAR ALGEBRA- 7 HOURS		
	LINEAR ALGEBRA- Vectors	1
2.1	VECTORS: geometry and algebra, Vector addition, and subtraction, Vector-scalar multiplication,	1
2.2	Dot product geometry, Vector orthogonality,	1
2.3	Vector Hadamard multiplication, cross product, unit vectors,	
2.4	VECTOR SPACES: Dimensions, and fields in linear algebra, Subspaces, Subspaces vs. subsets	1
2.5	Span, Linear independence, Basis	1
2.6	MATRICES: dimensionality, Matrix operations, Matrix-scalar multiplication,	1
2.7	Complex matrices, Addition, equality, transpose, Diagonal, and trace,	1
MODULE 3 -MATRICES – 7 HOURS		
3.1	MATRIX MULTIPLICATION: matrix multiplication by layering,	1

	Multiplication with diagonals, Matrix-vector multiplication,	
3.2	Symmetric matrices, multiply symmetric matrices Hadamard Multiplication, asymmetry Index, Code challenge,	1
3.3	RANK -concepts, Maximum possible rank, Computing rank, Rank and scalar multiplication, Rank of added and multiplied matrices	1
3.4	The rank of A & A^T , $A^T A$, AA^T , random matrices,	1
3.5	boosting rank by shifting, rank difficulties, rank, and span, Code challenge:	1
3.6	MATRIX SPACES: Column space and Row space of a matrix (A & AA^T), Null space of a matrix,	1
3.7	orthogonal subspaces, Dimensions of column/row/null spaces, Example of the four subspaces, code challenge,	1
MODULE 4- DETERMINANTS, & PROJECTIONS, AND ORTHOGONALIZATION- 10 HOURS		
4.1	DETERMINANT- Determinant of a 2x2 matrix, & 3x3 matrix, characteristic polynomial, the full procedure,	1
4.2	determinant of triangles, determinant and row reduction, determinant and scalar multiplication, theory vs practice, Code challenge	1
4.3	MATRIX INVERSE: Concept and applications, Inverse of a Diagonal matrix, Inverse of a 2x2 matrix, The MCA algorithm to compute the inverse,	1
4.4	Computing the inverse via row, Left inverse and right inverse, Pseudo-inverse, Code challenge	1
PROJECTIONS AND ORTHOGONALIZATION:		
4.5	Projections in R^2 , Projections in R^N , Orthogonal and parallel vector components,	1
4.6	Orthogonal matrices, Gram-Schmidt procedure, QR decomposition,	1
4.7	Inverse via QR Decomposition, Code challenge	1
LEAST SQUARES FOR MODEL-FITTING IN STATISTICS		
4.8	Introduction, least squares via left inverse, Least squares via orthogonal projection,	1
4.9	Least squares via row-reduction, Model-predicted values, and residuals	1
4.10	Least-squares applications, Code challenge	1
MODULE 5- DIMENSIONALITY REDUCTION- 8 HOURS		
EIGEN DECOMPOSITION-,		
5.1	Eigenvalues, eigenvectors, Eigen decomposition,	1
5.2	Diagonalization, Matrix powers via diagonalization	1
5.3	Distinct and repeated eigenvalues, symmetric matrices, Eigen layers of a matrix,	1

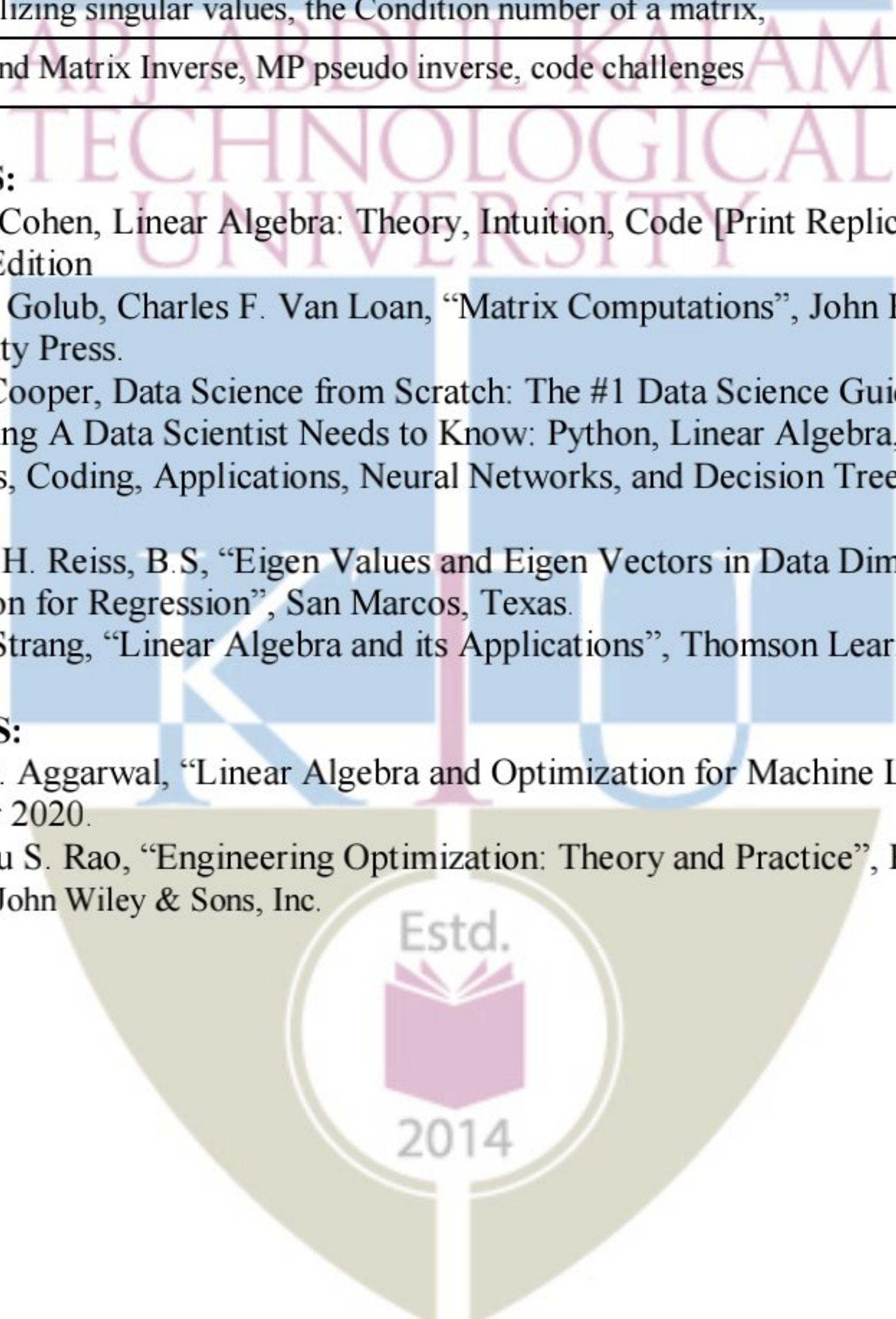
5.4	Eigen decomposition of singular matrices, Matrix powers and Inverse, Generalized eigen decomposition, Code challenges	1
SINGULAR VALUE DECOMPOSITION (SVD) -		
5.5	Singular value decomposition, Computing the SVD, singular values and eigenvalues,	1
5.6	Symmetric Matrices, SVD and the four subspaces, SVD, and matrix rank,	1
5.7	Spectral theory of matrices, SVD for low-rank approximations, Normalizing singular values, the Condition number of a matrix,	1
5.8	SVD and Matrix Inverse, MP pseudo inverse, code challenges	1

TEXT BOOKS:

1. Mike X Cohen, Linear Algebra: Theory, Intuition, Code [Print Replica] Kindle Edition
2. Gene H. Golub, Charles F. Van Loan, "Matrix Computations", John Hopkins University Press.
3. Steven Cooper, Data Science from Scratch: The #1 Data Science Guide for Everything A Data Scientist Needs to Know: Python, Linear Algebra, Statistics, Coding, Applications, Neural Networks, and Decision Tree Kindle Edition
4. Randolph H. Reiss, B.S, "Eigen Values and Eigen Vectors in Data Dimension Reduction for Regression", San Marcos, Texas.
5. Gilbert Strang, "Linear Algebra and its Applications", Thomson Learning Inc.

REFERENCES:

1. Charu C. Aggarwal, "Linear Algebra and Optimization for Machine Learning", Springer 2020.
2. Singiresu S. Rao, "Engineering Optimization: Theory and Practice", Fourth Edition 2009 by John Wiley & Sons, Inc.



221TCS004	INTRODUCTION TO AI AND NLP	CATEGORY	L	T	P	CREDIT
		Program Core 2	3	0	0	3

Preamble:

This course introduces the concepts, tools, and techniques of machine learning for text data. The students will learn the elementary concepts as well as emerging trends in the field of NLP. This course helps the learners to extract information from unstructured text, identify linguistic structure of it, and to apply different the techniques for text analytics. The learners will able to implement and evaluate NLP applications using machine learning and deep learning methods.

Course Outcomes:

After the completion of the course, the student will be able to: *

CO1	Analyze the applications of AI in the domain of NLP (Cognitive Knowledge Level: Apply)
CO2	Transform text into an appropriate data structure (Cognitive Knowledge Level: Apply)
CO3	Apply Probability models, language models, and Markov models for Text processing (Cognitive Knowledge Level: Apply)
CO4	Build NLP applications using Machine Learning Methods (Cognitive Knowledge Level: Apply)
CO5	Design, Develop, Implement and Present innovative Ideas on NLP and AI (Cognitive Knowledge Level: Create)

Program Outcomes (PO)

Outcomes are the attributes that are to be demonstrated by a graduate after completing the course.

PO1: An ability to independently carry out research/investigation and development work in engineering and allied streams

PO2: An ability to communicate effectively, write and present technical reports on complex engineering activities by interacting with the engineering fraternity and with society at large.

PO3: An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program

PO4: An ability to apply stream knowledge to design or develop solutions for real-world problems by following the standards

PO5: An ability to identify, select and apply appropriate techniques, resources and state-of-the-art tools to model, analyze and solve practical engineering problems.

PO6: An ability to engage in life-long learning for the design and development related to the stream related problems taking into consideration sustainability, societal, ethical and environmental aspects

PO7: An ability to develop cognitive load management skills related to project management and finance which focus on Entrepreneurship and Industry relevance.

Mapping of course outcomes with program outcomes

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1							
CO 2							
CO 3							
CO 4							
CO 5							

Assessment Pattern

Bloom's Category	End Semester Examination
Apply	50-80%
Analyse	20-40%

Evaluate	Assess using Assignments/Project
Create	Assess using Assignments/Project

Mark distribution

Total Marks	CIE	ESE	ESE Duration
100	40	60	2.5 hours

Continuous Internal Evaluation Pattern:

The evaluation shall only be based on application, analysis or design-based questions (for both internal and end-semester examinations).

Continuous Internal Evaluation: 40 marks

Micro project/Course based project : 20 marks

Course based task/Seminar/Quiz : 10 marks

Test paper, 1 no. : 10 marks

The project shall be done individually. Group projects are not permitted. Test paper shall include a minimum of 80% of the syllabus.

Course-based task/test paper questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation, and understanding of the students.

End Semester Examination Pattern:

The end-semester examination will be conducted by the University. There will be two parts; Part A and Part B. Part A contain 5 numerical questions with 1 question from each module, having 5 marks for each question. (Such questions shall be useful in the testing of knowledge, skills, comprehension, application, analysis, synthesis, evaluation, and understanding of the students). Students shall answer all questions.

Part B will contain 7 questions (such questions shall be useful in the testing of overall achievement and maturity of the students in a course, through long answer questions

Model Question Paper

QP CODE:

Reg No: _____

Name: _____

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
FIRST SEMESTER M.TECH DEGREE EXAMINATION, MONTH & YEAR

Course Code: 221TCS004

Course Name: INTRODUCTION TO AI AND NLP

Max. Marks: 60

Duration: 2.5 Hours

PART A

Answer All Questions. Each Question Carries 5 Marks

1. a. Contrast NLTK and Spacy 5
b. Explain the different types of artificial Intelligence
 2. a. Analyse the importance of the concept of vectors in NLP 5
b. Explain the simple functioning of the ELIZA chatbot
 3. Explain Markov Property and Markov model for NLP 5
 4. Discuss topic modeling and the intuition behind using LDA for topic modeling 5
 5. Describe implementing binary text classification using TensorFlow 5
- (5x5=25)**

Part B

(Answer any five questions. Each question carries 7 marks)

6. (a) Explain the applications of Artificial intelligence in Health care (7)

relating to theoretical/practical knowledge, derivations, problem-solving and quantitative evaluation), with a minimum one question from each module of which student should answer any five. Each question can carry 7 marks.

The total duration of the examination will be 150 minutes.

Course Level Assessment Questions

Course Outcome 1 (CO1):

1. Implement artificial intelligence-based solution in agriculture for optimization of irrigation and application of pesticides and herbicides
2. Develop Artificial Intelligence/Machine Learning based for Diabetes Care
3. Explain the role of Natural Language Processing Applications in Finance

Course Outcome 2 (CO2)

1. Which of the following techniques can be used for keyword normalization in NLP, the process of converting a keyword into its base form?

- a. Lemmatization
- b. Soundex
- c. Cosine Similarity
- d. N-grams

Interpret your answer.

2. Explain any two out of the eight methods that **Convert Text to Features**

- a. Recipe 1. One Hot encoding.
- b. Recipe 2. Count vectorizer.
- c. Recipe 3. N-grams.
- d. Recipe 4. Co-occurrence matrix.
- e. Recipe 5. Hash vectorizer.
- f. Recipe 6. Term Frequency-Inverse Document Frequency (TF-IDF)

g. Recipe 7. Word embedding.

h. Recipe 8. Implementing fast Text.

3. Explain the following with an example

a. Regular expression and its working

b. Properties of regular expression

c. Meta characters Big brackets [] and search [abcd] in 'kasdfaiabcasdfaabc' and write the output

d. Let's look at the following sentence: "I ate an apple and played the piano." Generate the one-hot embedding matrix for this sentence

Course Outcome 3(CO3):

- a. Implement TF-IDF from scratch
- b. Explain how you will implement NLP in other languages
- c. Explain Markov Model for text classifier and build a text classifier using the Markov model

Example: Take two sets of poems by two different authors, Edgar Allan Poe and Robert Frost. Given a sentence, the system should be able to predict the author

Course Outcome 4 (CO4):

1. Application: Latent Semantic Indexing for Search Engine Optimization using PCA/SVD
2. Implement ANN for multiclass classification
3. Implement Spam Detection using Naïve Bayes or Logistic regression

Course Outcome 5 (CO5):

1. Implement Text Summarization using python
2. How will you implement a sentiment analyzer in Python using logistic regression to predict sentiment on Amazon reviews?
3. Application: Topic Modeling Using Latent Dirichlet Allocation

7. (a) Elaborate vectors, their usefulness in NLP, tokenization, stop words, stemming, and lemmatization with examples (3)
- (b) Explain the limitations of the count vectorizer and the necessity of TF-IDF (2)
- (c) In a corpus of N documents, one randomly chosen document contains a total of T terms, and the term "hello" appears K times. What is the correct value for the product of TF (term frequency) and IDF (inverse-document-frequency), if the term "hello" appears in approximately one-third of the total documents? (2)
8. (a) Explain the regular expression and need for a regular expression in NLP (3)
- (b) How do regular expressions work? (2)
- (c) Explain the use of common Regex functions used in NLP (2)
9. (a) Explain the language model, applications of language modeling, method to Compute the probability of a sentence, and the curse of dimensionality (2)
- (b) what is Markov's assumption in language modeling and n-grams (3)
- (c) Implement n-grams and update-function (2)
10. (a) Explain the spam detection problem and why you want to detect them (2) (3)
- (b) Apply the Markov model n-gram approach to solve this problem and implement in Python (4)
11. (a) Explain RNN for text classification in TensorFlow (2)
- (b) Explain Parts of Speech (PoS) Tagging in TensorFlow (3)
- (c) Explain Named Entity Recognition in TensorFlow (2)
12. (a) Explain CNN for text classification (7)

Syllabus

Module	Contents	Total Lecture Hours (40 hrs)
1	INTRODUCTION TO ARTIFICIAL INTELLIGENCE: Artificial Intelligence? History, AI on a conceptual level, Types of AI, Use Cases, importance and applications of AI, AI algorithms, types of machine learning, types of problems solved in AI, advantages, and disadvantages of AI, AI In Marketing, Banking, Finance, Agriculture, HealthCare, Gaming, Space Exploration., Autonomous Vehicles, Chatbots, Artificial Creativity, AI Tools & Frameworks, AI vs Machine Learning vs Deep Learning, an overview of python for AI, INTRODUCTION TO NLP: NLP in the Real World, NLP Tasks, Language? Its Building Blocks, Why Is NLP Challenging? Machine Learning, Deep Learning, and NLP: An Overview, Approaches to NLP, Heuristics-Based NLP, Machine Learning & Deep Learning for NLP, NLP Pipeline, Applications of NLP-Machine translation, Speech recognition, Image Captioning, spam detection, text prediction- Introduction to Software Packages-Spacy, NLTK, Gensim, PyTorch, Regular Expression-importance, properties, working and python package (re), case study: working of Eliza chatbot	7
2	REGULAR EXPRESSION & TEXT PROCESSING: Common regex function, Meta Characters- Big brackets, cap, Backslash, Squared Brackets, Special Sequences, Asterisk, Plus, And Question mark, Curly Brackets Understanding Pattern Objects- Match Method Vs Search Method, Finditer Method, Logical Or, Beginning And End Patterns, Parenthesis String Modification- split method, sub-method, subn method, Text Processing-Words, Tokens, Counting words, vocabulary, corpus, tokenization in spacy- Sentiment Classification- (yelp) download a review dataset use data preparation using NumPy, pandas, counter, re-add tokens to vocabulary, build vocabulary from a data frame, from corpus, one hot encoding, encoding documents, train test splits, feature computation, confusion matrix, analysis. Language Independent Tokenization: Types of tokenization — Word, Character, and sub-word tokenization, problems with word tokenizer, drawbacks of a character-based tokenizer, problems with sub-word tokenization, Byte Pair Encoding, , String Matching and Spelling Correction-Minimum edit distance- table filling, dynamic programming,	9
	WORD EMBEDDING & PROBABILISTIC MODELS: Vector Models & Text Preprocessing: Vectors, Bag of Words, Count Vectorizer, Tokenization, Stopwords, Stemming, and Lemmatization, Stemming, and Lemmatization, Count Vectorizer, Vector Similarity. TF-IDF, Word-to-Index Mapping, Building TF-IDF, Neural Word Embeddings, Neural Word Embeddings. Vector Models Text Pre-processing Summary, steps of NLP analysis, Probabilistic Models-Language	9

3	Modelling: importance, types of language modeling, the curse of dimensionality, Language Model Markov Assumption And N-Grams, Language Model Implementation – Setup, Ngrams Function, Update Counts Function, Probability Model Function, Reading Corpus, Language Model Implementation Sampling Text, Markov Models: Markov Property, Markov Model, Probability Smoothing and Log-Probabilities, Building a Text Classifier, Article Spinning – Problem, N-Gram Approach, implementation, Cipher Decryption with Language Modeling And Genetic Algorithm Ciphers, substitution cipher, bigrams, maximum likelihood, and log-likelihood, Language models, Genetic Algorithms,	
4	NLP USING MACHINE LEARNING MODELS-Spam Detection– Problem, Naive Bayes theorem, Intuition, spam detection using Naïve Bayes, class imbalance, ROC, AUC, AND F1 SCORE, Implementing spam detection in python, Sentiment Analysis -Problem, Logistic Regression Intuition, Multiclass Logistic Regression, Logistic Regression Training and Interpretation, sentiment analysis implementation in python, Text Summarization-Using Vectors, Text Rank Intuition,Text Rank in Python, Text Summarization in Python Topic Modeling-different topic modeling techniques, Latent Dirichlet Allocation (LDA) – Essentials, Latent Dirichlet Allocation–Topic Modeling with Latent Dirichlet, Latent Symmetric Modelling(Indexing)-LSA / LSI Introduction, Singular Value Decomposition Intuition, LSA / LSI: Applying SVD to NLP, Latent Semantic Analysis / Latent Semantic Indexing in Python	7
5	DEEP LEARNING- word embeddings, nonlinear neural networks, Neuron – Intro, Fitting a Line, Classification Code Preparation, Text Classification in Tensorflow, The Neuron, How does a model learn?, Feed Forward Neural Networks- Ann-introduction, The Geometrical Picture, Activation Functions, Multiclass Classification, Text Classification ANN in Tensorflow, Text Preprocessing Code Preparation, Text Preprocessing in Tensorflow, Embeddings, CBOW(continuous bag of words), CBOW in Tensorflow, Convolution Neural Networks- Convolution, pattern matching, weight sharing, convolution in color images, CNN Architecture, CNN for Text, CNN for NLP in Tensorflow, Recurrent Neural Networks- Simple RNN / Elman Unit, RNNs: Paying Attention to Shapes, GRU, and LSTM. RNN for Text Classification in TensorFlow, Parts-of-Speech Tagging, and Named Entity Recognition in TensorFlow	8

COURSE PLAN		
Sl.NO	TOPIC	NO. OF LECTURES
	Module 1 – Introduction of AI & NLP (7 hours)	
Module 1: Introduction To Artificial Intelligence (7 hours)		
1.1	Artificial Intelligence? History, AI on a conceptual level, Types of AI, Use Cases, importance and applications of AI,	1
1.2	AI algorithms, types of machine learning, types of problems solved in AI, advantages, and disadvantages of AI	1
1.3	AI In Marketing, Banking, Finance, Agriculture, HealthCare, Gaming, Space Exploration, Autonomous Vehicles, Chatbots,	1
1.4	Artificial Creativity, AI Tools & Frameworks, AI vs Machine Learning vs Deep Learning, an overview of python for AI,	1
INTRODUCTION TO NLP,		
1.5	NLP in the Real World, NLP Tasks, Language? Its Building Blocks, Why Is NLP Challenging? Machine Learning, Deep Learning, and NLP: An Overview, Approaches to NLP, Heuristics-Based NLP, Machine Learning & Deep Learning for NLP	1
1.6	NLP Pipeline, Applications of NLP-Machine translation, Speech recognition, Image Captioning, spam detection, text prediction-	1
1.7	Introduction to Software Packages-Spacy, NLTK, Gensim, PyTorch, Regular Expression - importance, properties, working and python package (re), case study: working of Eliza chatbot	1
Module 2- Regular Expression & Text Processing (9 Hours)		
2.1	Common regex function used in NLP, -	1
2.2	Meta Characters- Big brackets, cap, Backslash,	1
2.3	Squared Brackets, Special Sequences, Asterisk, Plus, And Question mark, Curly Brackets	1
2.4	Understanding Pattern Objects - Match Method Vs Search Method, Finditer Method, Logical Or, Beginning And End Patterns, Parenthesis	1
2.5	String Modification - split method, sub-method, subn method,	1
TEXT PROCESSING		
2.6	Words, Tokens, Counting words, vocabulary, corpus, tokenization in spacy-	1
2.7	Sentiment Classification - (yelp) download a review dataset use –data preparation using NumPy, pandas, counter, re-add tokens to vocabulary, build vocabulary from a data frame, from corpus, one hot encoding, encoding documents, train test splits, feature computation, confusion matrix, analysis.	1

2.8	Language Independent Tokenization: Types of tokenization — Word, Character, and sub-word tokenization, problems with word tokenizer, drawbacks of a character-based tokenizer, problems with sub-word tokenization, Byte Pair Encoding	1
2.9	String Matching and Spelling Correction- Minimum edit distance- table filling, dynamic programming,	1
Module 3-Word Embedding & Probabilistic Models (9 Hours)		
3.1	Vector Models & Text Preprocessing: Vectors, Bag of Words, Count Vectorizer, Tokenization, Stop words,	1
3.2	Stemming and Lemmatization, Count Vectorizer, Vector Similarity. TF-IDF,	1
3.3	Word-to-Index Mapping, Building TF-IDF	1
3.4	Neural Word Embeddings, Neural Word Embeddings Demo. Vector Models & Text Pre-processing Summary, steps of a typical NLP analysis	1
	PROBABLISTIC MODELS	
3.5	Language Modelling: types of language modeling, the importance of language modeling, the curse of dimensionality, Language Model Markov Assumption And N-Grams,	1
3.6	Language Model Implementation – Setup, Ngrams Function, Update Counts Function, Probability Model Function, Reading Corpus, Language Model Implementation Sampling Text,	1
3.7	Markov Models: Markov Property, Markov Model, Probability Smoothing and Log-Probabilities, Building a Text Classifier,	1
3.8	Article Spinning –Problem Description, N-Gram Approach, implementation in python,	1
3.9	Cipher Decryption with Language Modeling And Genetic Algorithm- Ciphers, substitution cipher, bigrams, maximum likelihood, and log-likelihood, Language models, Genetic Algorithms,	1
Module 4 – NLP Using Machine Learning Models (7 Hours)		
4.1	Spam Detection – Problem, Naive Bayes theorem, Intuition, spam detection using Naïve Bayes, class imbalance, ROC, AUC, AND F1 SCORE, Implementing spam detection in python,	1
4.2	Sentiment Analysis -Problem, Logistic Regression Intuition, Multiclass Logistic Regression,	1
4.3	Logistic Regression Training and Interpretation, sentiment analysis implementation in python	1
4.4	Text Summarization -Using Vectors, Text Rank Intuition	1
4.5	Text Rank in Python, Text Summarization in Python	1
4.6	Topic Modeling -different topic modeling techniques, Latent Dirichlet Allocation (LDA) – Essentials, Latent Dirichlet Allocation–Topic Modeling with Latent Dirichlet	1
4.7	Latent Symmatc Modelling(Indexing) -LSA / LSI Introduction, Singular Value Decomposition Intuition, LSA / LSI: Applying SVD to NLP, Latent Semantic Analysis / Latent Semantic Indexing in Python	1

Module 5 - Deep Learning (8 Hours)		
5.1	word embeddings, nonlinear neural networks	1
5.2	Neuron – Intro, Fitting a Line, Classification Code Preparation, Text Classification in Tensorflow, The Neuron, How does a model learn?,	1
5.3	Feed Forward Neural Networks- Ann-introduction, The Geometrical Picture, Activation Functions, Multiclass Classification, Text Classification ANN in Tensorflow,	1
5.4	Text Preprocessing Code Preparation, Text Preprocessing in Tensorflow, Embeddings, CBOW(continuous bag of words), CBOW in Tensorflow	1
5.5	CONVOLUTION NEURAL NETWORKS:- CNN-Introduction, Convolution, pattern matching, weight sharing, convolution in color images,	1
5.6	Convolution Neural Networks- Convolution, pattern matching, weight sharing, convolution in color images, CNN Architecture, CNN for Text, CNN for NLP in Tensorflow,	1
5.7	Recurrent Neural Networks- Simple RNN / Elman Unit, RNNs: Paying Attention to Shapes, , GRU, and LSTM. RNN for Text Classification in TensorFlow,	1
5.8	Parts-of-Speech Tagging, and Named Entity Recognition in TensorFlow	1

Reference Books

1. Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, Practical Natural Language Processing, Shroff/O'Reilly, 2020.
2. Steven Bird, Ewan Klein, Edward Loper, Natural Language Processing with Python, O'Reilly
3. Akshay Kulkarni, Adarsha Shivananda, Natural Language Processing Recipes Unlocking Text Data with Machine Learning and Deep Learning using Python, Apress
4. Taweh Beysolow II 'Applied Natural Language Processing with Python- Implementing Machine Learning and Deep Learning Algorithms for Natural Language Processing, Apress
5. Palash Goyal, Sumit Pandey, Karan Jain 'Deep Learning for Natural Language Processing Creating Neural Networks with Python, Apress
6. Hobson Lane. Cole Howard, Hannes Max Hapke 'Natural Language Processing in Action, Understanding, analyzing, and generating text with Python', ©2019 by Manning Publications Co
8. Wolfgang Ertel, Introduction to Artificial Intelligence, Springer,